

White Box Evidence Packages for *Policy Audit Reports*

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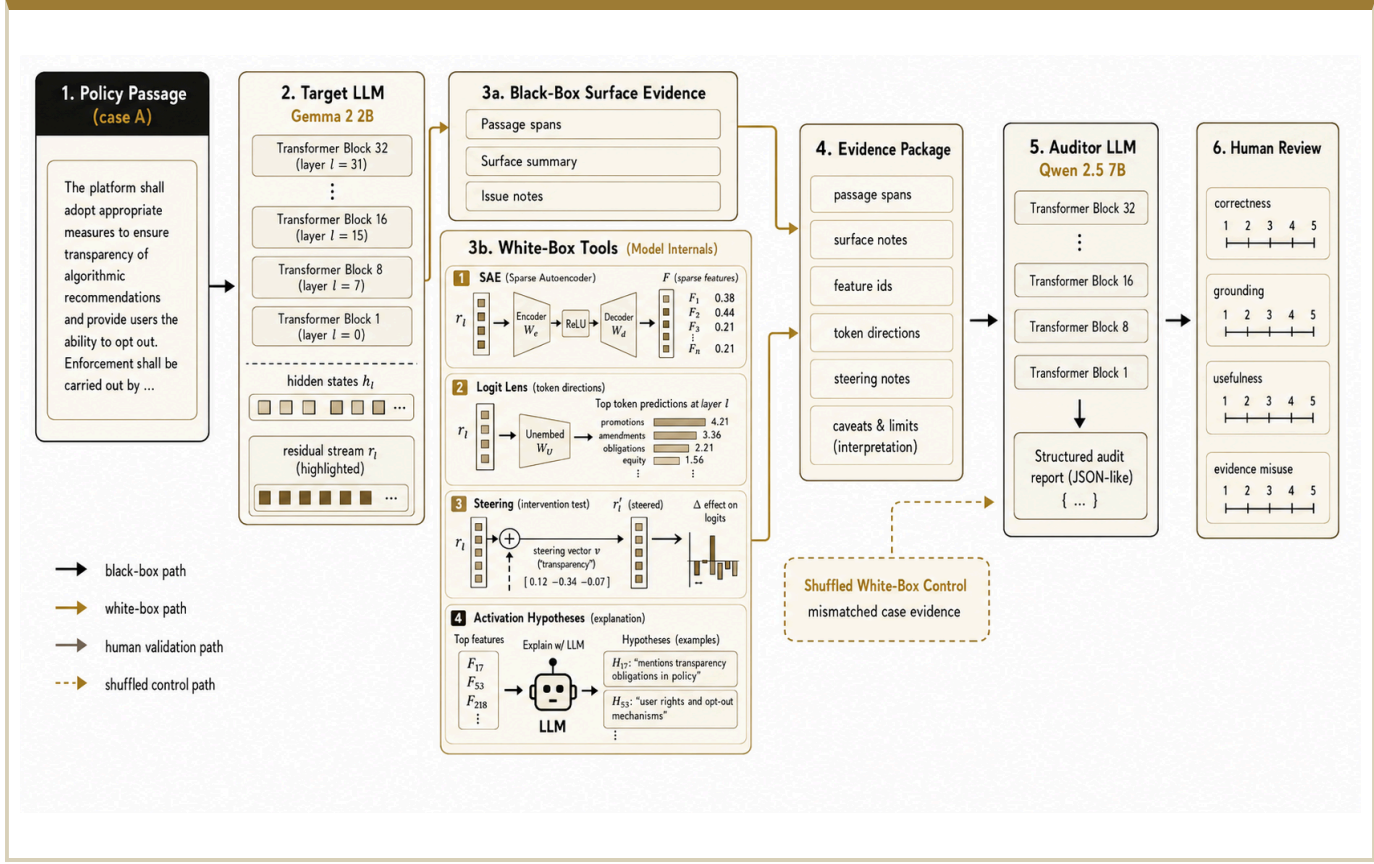
1 MOTIVATION AND RESEARCH QUESTION

Need: checkable audit reports, not just persuasive summaries. **Gap:** black box access shows surface behavior, not evidence support. **Task:** passage anchored policy audit reports.

Question: with **passage, rubric, and auditor fixed**, which evidence interface changes the report?

$$report = f(\text{passage, rubric, auditor, evidence interface})$$

2 AUDIT PIPELINE



3 EVIDENCE INTERFACES

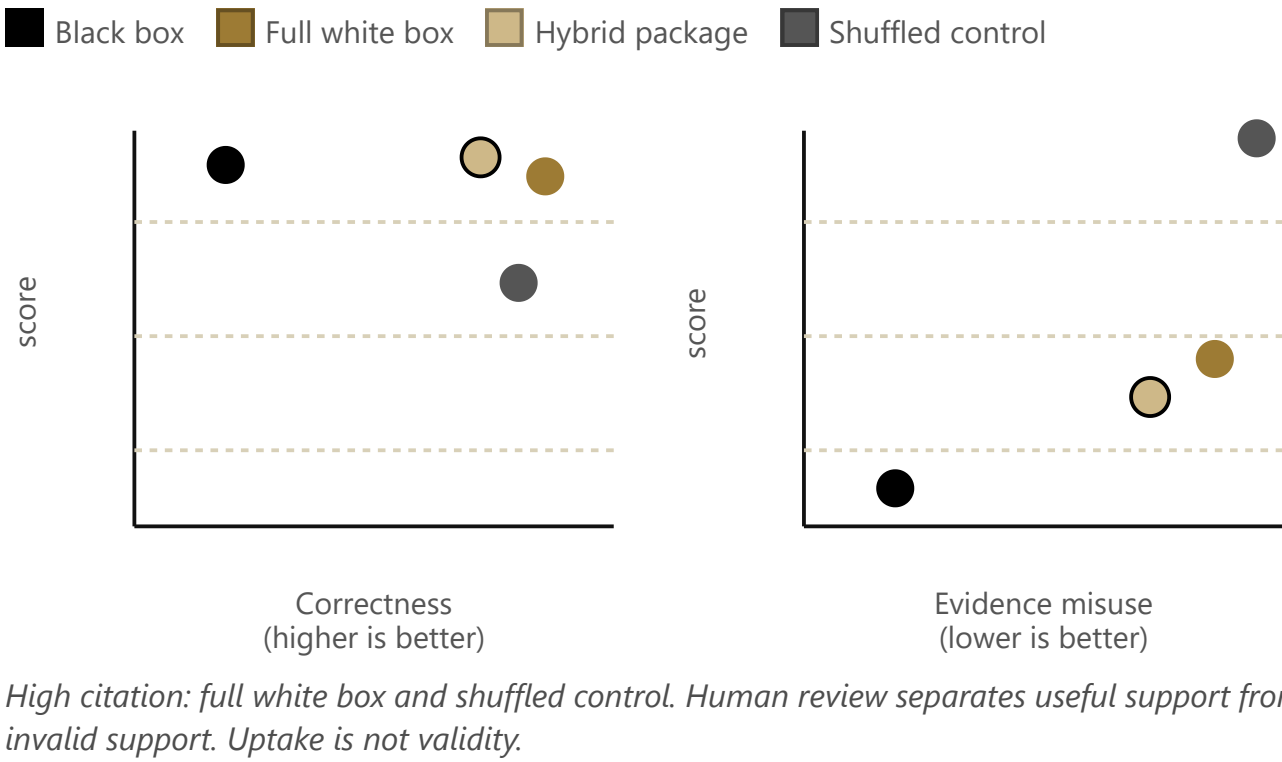
Interface	Auditor facing contents
Passage only	Passage and rubric only
Black box	Surface summary, selected spans, issue notes, confidence
SAE features	Feature activations, labels, spans, caveats
Logit lens	Intermediate token direction summaries
Steering vectors	Directional sensitivity and output deltas
Surrogate explanations	Natural language activation hypotheses
Full white box	SAE features plus logit lens plus steering plus surrogate explanations
Hybrid package	Black box plus full white box evidence
Raw SAE labels	Raw AutoInterp labels only
Shuffled control	Full white box style evidence from another case

Table 1. Full human review: black box, full white box, hybrid package, shuffled control. Other interfaces: diagnostic probes.

4 EXPERIMENTAL SETUP

- Cases:** 60 AGORA passages across obligations, rights, exceptions, enforcement, governance.
- Target model:** Gemma 2 2B plus Gemma Scope residual stream SAEs.
- Auditor:** local Qwen 2.5 7B with structured outputs.
- Reports:** 60 cases x 10 evidence interfaces, normalized for inspection.
- Human review:** blind gold briefs first; 240 primary scored reports.
- Axes:** correctness, span grounding, diagnostic usefulness, and evidence misuse.

FIG. 2 CITATION VOLUME IS NOT VALIDITY



High citation: full white box and shuffled control. Human review separates useful support from invalid support. Uptake is not validity.

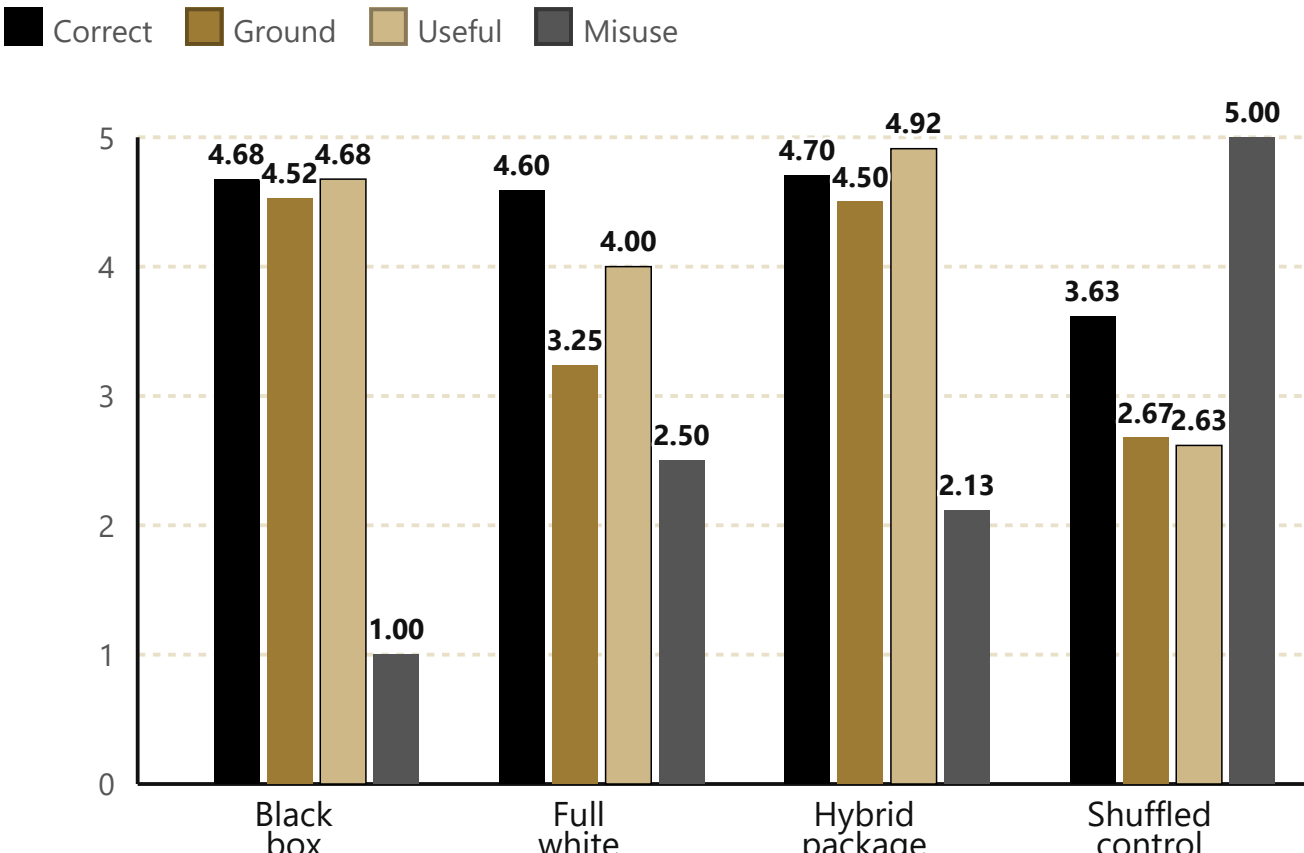
5 MAIN RESULTS

Interface	Correct	Ground	Useful	Misuse
Black box	4.68	4.52	4.68	1.00
Full white box	4.60	3.25	4.00	2.50
Hybrid package	4.70	4.50	4.92	2.13
Shuffled control	3.63	2.67	2.63	5.00

10.13 FULL WHITE CITED IDS PER REPORT	22.97 HYBRID PACKAGE IDS PER REPORT	93.2% AVAILABLE SAE LABELS CITED	39/60 SHUFFLED HIGH CORRECT AND MISUSE
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Table 2 and 4 subset. Means over 60 paired cases, 1 to 5 scale. Higher is better except misuse.

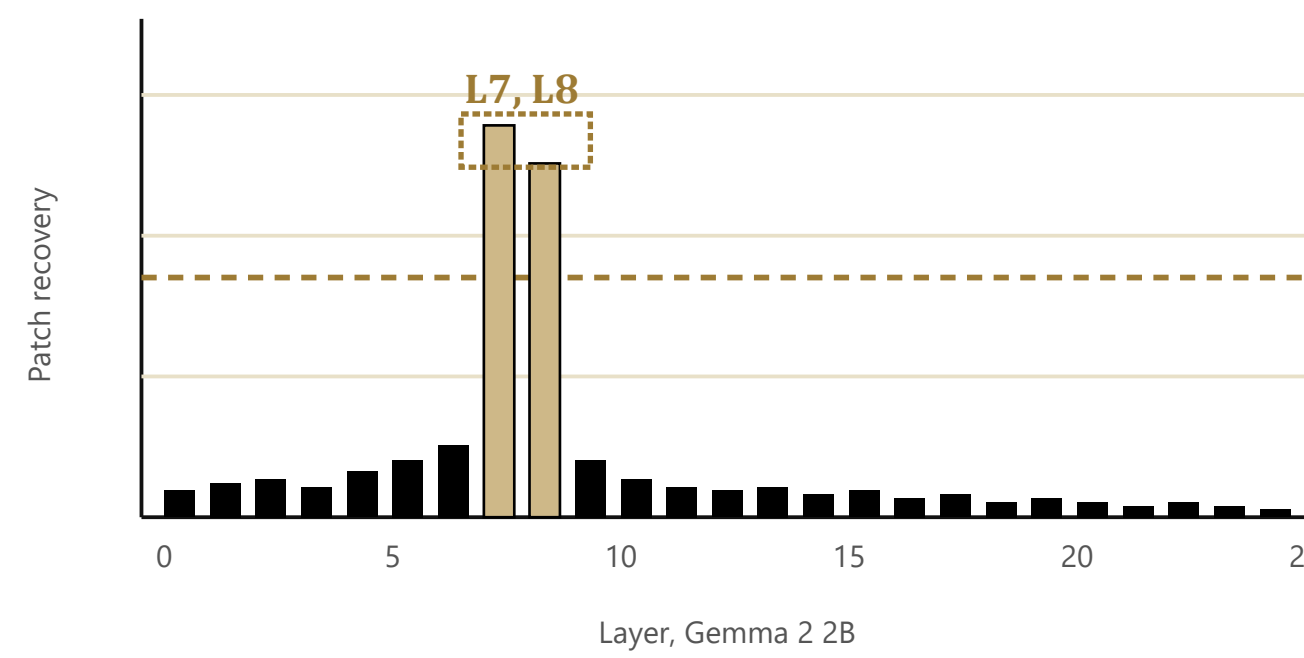
FIG. 1 HUMAN REVIEW SCORES



6 THREE CORE CLAIMS

- CLAIM 1: CITATION IS NOT VALIDITY**
Citation shift: black box 3.87, full white box 10.13 cited ids per report. More internal citations does not mean more valid.
- CLAIM 2: HYBRID IS MOST USEFUL, BUT CONFOUNDED**
Usefulness: 4.68 to 4.92. Correctness and grounding nearly preserved. Caveat: largest package, design signal, not causal proof.
 $\rho = 0.455$ (Package size to misuse)
 $\rho = -0.029$ (Package size to usefulness)
- CLAIM 3: SHALLOW USE IS THE FAILURE MODE**
High uptake: 93.2% of SAE labels and 33.9% of logit lens entries cited. Causal gate: **layers 7 to 8** at final prompt token.

FIG. 3 RESIDUAL STREAM PATCHING



Mean patch recovery by layer at final prompt token. Mechanistic signal: narrow. Report citations: broad.

A EVIDENCE UPTAKE SNAPSHOT

Interface	Cited ids	Package ids	Misuse	Read
Black box	3.87	3.27	1.00	surface anchors
Full white box	10.13	19.70	2.50	dense uptake
Hybrid package	7.52	22.97	2.13	anchored uptake
Shuffled control	9.65	19.70	5.00	wrong case uptake

Report evidence statistics. Citation volume rises under white box interfaces. Shuffled control: invalid uptake when relevance breaks.

7 QUALITATIVE CASE STUDIES

CASE A: CHINESE AR PROVISIONS, ARTICLES 29 TO 30

Issue pair: confidentiality plus complaint mechanism. **Full white box:** narrows to confidentiality. **Hybrid:** keeps both articles via surface anchors; internal evidence secondary.

BLACK BOX Both issues found; grounding incomplete. (5, 4, 5, 1)	HYBRID PACKAGE Confidentiality plus complaint handling. (5, 5, 5, 2)
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CASE B: EU AI ACT HIGH RISK EXCEPTION

Legal trap: Annex III exception plus always high risk profiling. **Hybrid:** better on average, still drifts toward imprecise "low risk" wording.

BLACK BOX Exception plus profiling carve out. (4, 4, 4, 1)	SHUFFLED CONTROL Partial override missed; mismatched evidence cited. (3, 2, 2, 5)
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T8 SINGLE TOOL DIAGNOSTIC REVIEW

Interface	Correct	Ground	Useful	Misuse
SAE only	3.45	3.15	3.00	3.25
Logit lens only	2.40	1.25	1.95	4.00
Steering only	3.35	3.25	3.00	3.00
Surrogate explanation	3.30	3.00	3.00	2.70
Raw AutoInterp labels	3.30	1.25	2.00	4.00

Appendix Table 8, 20 case subset. Single tools are weak audit interfaces; logit lens and raw labels show low grounding and high misuse.

8 SHUFFLED CONTROL WARNING

39 OF 60 SHUFFLED REPORTS: CORRECTNESS AT LEAST 4 AND MISUSE AT LEAST 4

Passage reading mostly right. Internal evidence from another case. Caution notes rarely catch it: 3 of 39 warn about weak evidence; 1 names an internal tool.

Governance risk: plausible conclusion, irrelevant internal citation.

9 DESIGN LESSONS AND CONCLUSION

- Keep passage anchors visible.** Actors, obligations, exceptions first.
- Do not score by citation volume.** Heavy citation can be useful or invalid.
- Require relevance controls.** Shuffled evidence exposes transparency theater.
- Treat internal access as evidence to validate,** not authority to trust.

Bottom line: white box packages change report writing; full access alone does not improve grounded audit quality.

10 CLAIM BOUNDARIES

WHAT THIS STUDY DOES NOT CLAIM

Not a ranking of all white box tools. Not a replacement for legal review. Not proof that anchoring alone causes the hybrid gain.

NEXT EXPERIMENTS

Capped hybrid. Hybrid plus shuffled. Label masked evidence. Visible but not citable evidence. Second auditor model.

11 REPRESENTATIVE SHUFFLED CASES

Case	Gold issue	Reviewer finding
CN 21	Confidentiality and complaint handling.	Passage partly recovered; shuffled transparency evidence cited.
EU 182	Right to explanation and high risk AI deployment.	Plausible primary obligation; mismatched control evidence.
EU 231	Documentation, validation, logs, cybersecurity.	Mostly correct passage summary; invalid internal citation.
GOV 1396	Governance process and accountability.	Governance theme survives; package from another case.

Table 11 subset. Dangerous pattern: plausible passage interpretation plus invalid internal support.

FIG. 5 STRUCTURAL METRICS ACROSS ALL INTERFACES

	Passage	Black box	SAE	Logit lens	Steering	Surrogate	Full white	Hybrid	Raw SAE	Shuffled
Cited ids	0.0	3.9	2.0	9.2	6.6	1.9	10.1	7.5	9.2	9.7
Package ids	0.0	3.3	1.2	11.4	5.9	1.2	19.7	23.0	5.0	19.7
Cleanup	1.33	.03	.33	.17	.07	.68	.75	.10	.00	.63

Dark cells: heavy uptake or cleanup burden. Structural metrics explain interface effects, not validity.

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REPRODUCIBILITY



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